INCH-POUND

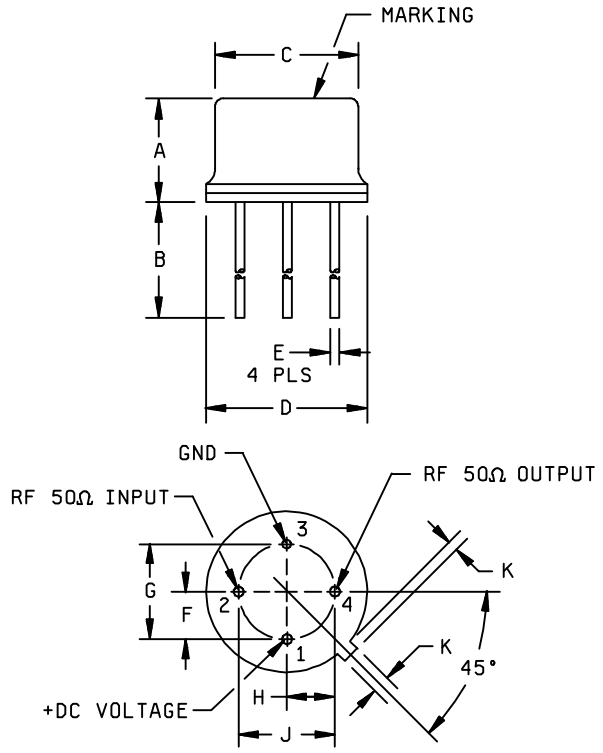
MIL-DTL-28875/8A
24 August 2000
SUPERSEDING
MIL-A-28875/8
27 July 1983

DETAIL SPECIFICATION SHEET

AMPLIFIERS, RADIO-FREQUENCY AND MICROWAVE, SOLID-STATE, POWER, WIDEBAND, TO CONFIGURATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the amplifiers described herein shall consist of this specification sheet and the latest revision of MIL-DTL-28875.



	Inches		Millimeters	
	Max	Min	Max	Min
B	N/A	.17	N/A	4.32
C	.455	.445	11.56	11.30
D	.510	.490	12.95	12.45
E	.019	.016	.48	.41
F	.160	.140	4.06	3.56
G	.310	.290	7.87	7.37
H	.160	.140	4.06	3.56
J	.310	.290	7.87	7.37
K	.05	N/A	1.3	N/A

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).

FIGURE 1. Dimensions and configuration.

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TABLE I. Electrical characteristics (25°C unless otherwise specified).

Dash No.	Freq. range	DC input power		Gain (dB)	Gain flatness (dB)	RF output power (dBm)	VSWR Max		Noise figure (dB)	Intercept point 3rd order (dBm)	Harmonic output (dBc)	Gain versus temp. (dB)	RF input power (mW)	Iso-lation (dB)	Phase linearity (deg)
	(MHz)	Voltage	Curr. Max (mA)	Min	Max	Min	IN	OUT	Max	Min			Max	Min	Max
S1 01 S2 N	10-400	+15 ±1%	35	12	±1.0	+10	2.0:1	2.0:1	7.5	+21	N/A	±1	32	N/A	N/A
S1 02 S2 N	2-500	+15 ±1%	65	12	±0.7	+14	2.0:1	2.0:1	7.0	+25	N/A	±1	N/A	N/A	N/A
S1 03 S2 N	5-500	+24 ±1%	50	9	±1.0	+13	2.0:1	2.0:1	7.0	+27	N/A	±1	N/A	N/A	N/A
S1 04 S2 N	5-500	+15 ±1%	80	23	±1.0	+12	2.0:1	2.0:1	7.0	+25	N/A	±1	N/A	N/A	N/A
S1 05 S2 N	5-500	+15 ±1%	48	15	±1.0	+13	2.0:1	2.0:1	6.0	+28	N/A	±1	N/A	N/A	N/A
S1 06 S2 N	5-500	+15 ±1%	50	16	±1.0	+17	2.5:1	2.5:1	6.0	+26	N/A	±1	N/A	N/A	N/A
S1 07 S2 N	10-500	+15 ±1%	121	14	±1.0	+22	2.0:1	2.0:1	9.0	+35	N/A	±1	N/A	N/A	N/A
S1 08 S2 N	10-500	+15 ±1%	105	9	±1.0	+22	2.0:1	2.0:1	9.0	+30	N/A	±1	N/A	N/A	N/A
S1 09 S2 N	10-1000	+15 ±1%	114	5.5	±1.3	+19	2.2:1	2.2:1	12.5	+34	N/A	±1	100	N/A	N/A

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TABLE I. Electrical characteristics (25°C unless otherwise specified) - Continued.

Dash No.	Freq. range	DC input power		Gain (dB)	Gain flatness (dB)	RF output power (dBm)	VSWR Max		Noise figure (dB)	Intercept point 3rd order (dBm)	Harmonic output (dBc)	Gain versus temp. (dB)	RF input power (mW)	Isolation (dB)	Phase Linear. (deg)
	(MHz)	Voltage	Curr. Max (mA)	Min	Max	Min	IN	OUT	Max	Min			Max	Min	Max
S110 S2N	10-1500	+15 ±1%	45	9.5	±1.0	+13.5	2.0:1	2.0:1	7.5	+29	N/A	±1	20	N/A	N/A
S111 S2N	10-1500	+15 ±1%	95	3	±1.2	+19	2.5:1	2.5:1	13.5	+32	N/A	±1	80	N/A	N/A

1/ Both RF input and output pins are at dc ground.
2/ RF input is at dc ground.

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TABLE II. Dimension, physical and environmental characteristics.

Dash number	Dimension A in inches (mm) Max	Mass Max (Grams)	Temperature range		Burn-in temperature (case temp)	Seal	Salt
			Operating case temp	Nonoperating			
01 S1 S2 N	.26 (6.6)	1.9	-55/+100°C	-65/+125°C	+125°C	Hermetic	N/A
02 S1 S2 N	.18 (4.6)	1.9	-54/+85°C	-55/+125°C	+100°C	Hermetic	N/A
03 S1 S2 N	.18 (4.6)	1.9	-54/+85°C	-55/+125°C	+100°C	Hermetic	N/A
04 S1 S2 N	.21 (5.3)	1.9	-65/+110°C	-65/+125°C	+110°C	Hermetic	N/A
05 S1 S2 N	.21 (5.3)	1.9	-54/+85°C	-62/+125°C	+100°C	Hermetic	N/A
06 S1 S2 N	.21 (5.3)	1.9	-55/+85°C	-62/+100°C	+95°C	Hermetic	N/A
07 S1 S2 N	.18 (4.6)	1.9	-54/+71°C	-62/+100°C	+71°C	Hermetic	N/A
08 S1 S2 N	.18 (4.6)	1.9	-54/+85°C	-62/+125°C	+100°C	Hermetic	N/A
09 S1 S2 N	.21 (5.3)	1.9	-54/+85°C	-62/+125°C	+100°C	Hermetic	N/A
10 S1 S2 N	.21 (5.3)	1.9	-54/+85°C	-62/+125°C	+100°C	Hermetic	N/A
11 S1 S2 N	.21 (5.3)	1.9	-54/+85°C	-62/+125°C	+95°C	Hermetic	N/A

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table II.

Electrical characteristics: See table I.

Physical and environmental characteristics: See table II.

Part number. M28875/8-(dash number from tables I and II).

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Custodian:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AV, MI
Navy - AS, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5985-1208-05)

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